Practice runs and evidence

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to prepare an evidence-based design explanation.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by repeated evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 5 times as large.
Compare ideal travel speed and state one real-world limitation.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent.
Retain every trial and required build evidence.

Worked reasoning: Supplied 3 m runs take 2.0, 2.4, 2.1 s. Report all trials; their mean time is 2.17 s. “Fastest run 2.0 s” answers a different question from typical performance.

Your turn: Make a trial table and mark both mean and fastest time.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

03 Look and explain

Complete a final labeled sketch or attach a photo with labels.
Identify a feature that supports straight motion and one that supports reliable electrical operation.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

04 Build the solution

A supplied 4 m practice course takes 5 s.
Find average speed. Then record your actual measured course length and three run times.

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Distance (m)	Time (s)
1		
2		
3		

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

One run stops early after a loose connection. Should you erase it?
Explain how to record it fairly and decide whether it belongs in a speed comparison.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A magnet moves through a 60-turn coil. Propose two controlled changes that could increase induced-emf magnitude and explain each.

Word help: emf = energy supplied per charge, measured in volts

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

Name the final design change you will keep and cite one specific observation or measurement supporting it.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2.
Compare this set with 2.1, 2.1, 2.1 s. Discuss speed and consistency rather than declaring one winner from the fastest run alone.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3